

1. Administrative & Study Identification

Official Study Title: MEMENTO - A Phase 4, Single-arm, Open-label Clinical Study to Evaluate Mavacamten in Adults With Symptomatic Obstructive Hypertrophic Cardiomyopathy to Assess the Impact on Myocardial Structure With Cardiac Magnetic Resonance Imaging (CMR) **Brief Study Title:** A Study to Evaluate Mavacamten Impact on Myocardial Structure in Participants With Symptomatic Obstructive Hypertrophic Cardiomyopathy **Protocol Number:** CV027-1088 **NCT Number:** NCT06112743 **Posting ID:** 979535842538204048 **Sponsor / Company Name:** Bristol Myers Squibb **Investigational Product:** Mavacamten (Trade Name: Camzyos) [ATC Code: C01EB24] **Phase of Development:** Phase 4 **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the impact of mavacamten on myocardial structure using cardiac magnetic resonance imaging (CMR) in adult participants with symptomatic obstructive hypertrophic cardiomyopathy (oHCM) [New York Heart Association (NYHA) Functional Class II or III]. **Secondary Objectives:** To evaluate the safety, efficacy, and tolerability of mavacamten, and its broader impact on functional capacity, symptoms, left ventricular outflow tract (LVOT) gradients, and biomarker levels (e.g., NT-proBNP).

2.2 Background & Rationale To address the need for disease-modifying therapies in symptomatic oHCM. Hypertrophic cardiomyopathy (HCM) causes thickened heart muscle that can obstruct blood flow (oHCM). Traditional therapies mostly palliate symptoms rather than targeting the underlying molecular defect. Mavacamten is a novel, first-in-class cardiac myosin inhibitor that reduces myocardial hypercontractility by targeting the sarcomere. This study aims to use serial CMR to capture real-world structural reverse remodeling data, highlighting the drug's potential to slow or halt disease progression.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** Non-Randomized

3.2 Planned Enrollment & Duration **Targeted Enrollment (\$N\$):** 63 participants **Number of Sites:** Multicenter / Global **Estimated**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Diagnosed with obstructive hypertrophic cardiomyopathy (oHCM), in accordance with current American College of Cardiology Foundation/American Heart Association and European Society of Cardiology guidelines. 3. Left ventricular outflow tract (LVOT) peak gradient ≥ 30 mmHg and ≥ 50 mmHg after Valsalva or after exercise. 4. Left ventricular ejection fraction (LVEF) $\geq 55\%$ at rest. 5. New York Heart Association (NYHA) functional class II or III symptoms. 6. Other protocol-defined inclusion criteria apply.

4.2 Exclusion Criteria 1. A known infiltrative or storage disorder causing cardiac hypertrophy that mimics oHCM. 2. Documented obstructive coronary artery disease or history of myocardial infarction. 3. A history of resuscitated sudden cardiac arrest or life-threatening ventricular arrhythmia within 6 months prior to screening. 4. An implantable cardioverter defibrillator (ICD) or pacemaker, or another contraindication for cardiac magnetic resonance imaging (CMR). 5. Other protocol-defined exclusion criteria apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Mavacamten specific dose based on standard titration protocols guided by echocardiography and LVOT gradient/LVEF assessments. **Route of Administration:** Oral **Duration of Treatment:** 48 weeks for the primary CMR assessment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard background background therapies such as β -blockers or calcium channel blockers. **Prohibited:** Disopyramide or any other interacting agents that strongly induce or inhibit CYP450 pathways, avoiding excessive negative inotropy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Echo, Initial CMR
Baseline	Day 0	Treatment Initiation, Vitals, Baseline Biomarkers	Interim
Weeks 4, 8, 12, 24, 36	Safety Labs, LVEF & LVOT Gradient Echocardiography, NYHA Assessment, Dose Titration	End of Study	Week 48 Final CMR Assessment, Primary Endpoint Evaluation, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Composite of maximum left atrial volume index (LAVI) and left ventricular mass index (LVMI) at Week 48. Defined as participants achieving *both* of the following criteria at Week 48 CMR assessment: * A decrease of at least 5 mL/m² in maximum LAVI from baseline * A decrease of at least 5 g/m² in LVMI from baseline **Measurement Method:** Cardiac Magnetic Resonance Imaging (CMR) with central laboratory adjudication.

7.2 Secondary & Exploratory Endpoints * Proportion of participants who had at least 1 class of improvement from baseline in New York Heart Association (NYHA) class at Week 48. * Change from baseline in maximum left atrial volume index (LAVI) at Week 48. * Change from baseline in left ventricular mass index (LVMI) at Week 48. * Change from baseline in resting and post-exercise Valsalva LVOT peak gradients. * Change from baseline in cardiac biomarkers (NT-proBNP and high-sensitivity cardiac troponin).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation via patient report and closely monitored serial echocardiography tracking for transient LVEF reduction (< 50%). **Serious Adverse Events (SAEs):** Reported within 24 hours (e.g., heart failure exacerbation, syncope, significant LVEF drop). **Data Monitoring Committee (DMC):** Independent safety monitoring to oversee drug-related systolic dysfunction and proper adherence to the dose-titration algorithm.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary CMR structural endpoints; Safety Set for tracking LVEF stability and adverse events. **Sample Size Justification:** Target enrollment of N=63 powered to detect structural differences via the highly sensitive CMR methodology. **Handling of Missing Data:** Standard multiple imputation for structural CMR endpoints missed due to premature trial discontinuation or technical imaging failure.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring specifically tracking LVEF safety parameters; strict adherence to the Mavacamten Risk Evaluation and Mitigation Strategy (REMS) program. **Audit Trail:** eCRF locking, centralized core lab

evaluations of baseline and 48-week CMR images to ensure unbiased assessment.

11. Study Identifiers & Governance

Primary ID: CV027-1088 **Registry Number:** NCT06112743 **Posting ID:** 979535842538204048 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate Mavacamten Impact on Myocardial Structure in Participants With Symptomatic Obstructive Hypertrophic Cardiomyopathy **Official Regulatory Title:** MEMENTO - A Phase 4, Single-arm, Open-label Clinical Study to Evaluate Mavacamten in Adults With Symptomatic Obstructive Hypertrophic Cardiomyopathy to Assess the Impact on Myocardial Structure With Cardiac Magnetic Resonance Imaging (CMR)

12. Clinical Framework (oHCM Specific)

Disease Areas / Conditions: Cardiomyopathy, Hypertrophic
Preferred UMLS Name: Obstructive Hypertrophic Cardiomyopathy
Primary Condition: Symptomatic Obstructive Hypertrophic Cardiomyopathy (oHCM) **Semantic Type:** Functional Concept **SNOMED ID:** 56246009 **Diagnosis Threshold:** LVOT peak gradient ≥ 30 mmHg (rest) and ≥ 50 mmHg (provoked); LVEF $\geq 55\%$ **Medical Methodology:** Cardiac Myosin Inhibition (Sarcomere targeted)
Optimization Target: Reverse structural remodeling defined by reduction in LAVI and LVMI

13. Study Procedures & Methodology

Management Pathway: Structural monitoring via serial CMR and safety monitoring via clinical echocardiography **Education Protocol (HCP):** Mandatory training on mavacamten titration, monitoring for systolic dysfunction, and heart failure symptoms. **Education Protocol (Patient):** Counseling on strict adherence to scheduled echocardiograms and prompt reporting of new-onset shortness of breath or edema. **Quality Audit Mechanism:** Blinded Central Core Lab readings for all CMR outcomes to eliminate bias in open-label design.

14. Observation & Follow-Up Schedule

Total Duration: 48 Weeks (Primary Endpoint Timeframe) **Onsite Visit Cadence:** Frequent titration visits initially (e.g., Weeks 4, 8, 12), with imaging milestones up to Week 48. **Remote Monitoring Frequency:** Between clinic visits as necessitated by the REMS program. **Checklist Review Interval:** Routine symptom and NYHA functional class evaluation every 12 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					